أسئلة سنوات

فاينل

لجنة سنافر البوليتكنك - بسواعدنا نبنيها

اسم المادة

الكيمياء العامة

نبنيها بسواعدنا

تواصل معنا

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Muslimengineerpage

96500 C/mole e⁻

Transfer your answer to the table below in CAPITAL LETTERS

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer					D	E	C						

Question	14	15	16	17	18	19	20	21	22	23	24	25
Answer									E		E	E

Choose the most correct answer for each of the following questions:

1. Consider the unbalanced redox equation:
$_CH_3OH_{(l)} + _Cr_2O_7^{2-}{}_{(aq)} + _H^+{}_{(aq)} \rightarrow __CH_2O_{(aq)} + _Cr^{3+}{}_{(aq)} + __H_2O_{(l)}$. Which of the following sets of numbers will balance the equation?
A) 3, 1, 8, 3, 2, 7 B) 1, 3, 8, 3, 2, 4 C) 8, 1, 3, 4, 3, 2 D) 3, 8, 1, 2, 3, 7
E) 3, 1, 8, 3, 2, 4

2. In which of the following does sulfur have an oxidation number of +7?
A) H_2SO_3 B) $H_2S_2O_3$ C) H_2SO_4 D) $H_2S_2O_8$ E) S_2F_{10}

3. Examine the data on the right. The strongest oxidizing agent among these substances is:
A) Sr^{2+} B) Cr^{2+} C) Co^{2+}
D) Zn^{2+} E) Fe^{2+}

half-reduction reactions	E° (V)
$Sr^{2+} + 2e^- \rightarrow Sr$	- 2.89
$Cr^{2+} + 2e^- \rightarrow Cr$	- 0.913
$Fe^{2+} + 2e^- \rightarrow Fe$	- 0.447
$Co^{2+} + 2e^- \rightarrow Co$	- 0.280
$Zn^{2+} + 2e^- \rightarrow Zn$	- 0.760

4. The mass (in g) of Cu (63.55 g/mol) that can be deposited by the passage of 12.0 A for 25.0 min through $CuSO_{4(aq)}$ solution equals:
A) 5.93 B) 16.59 C) 6.91 D) 19.36 E) 14.38

5. A flask containing 0.124 mol of A and was allowed to react according to the equation $A_{(g)} \rightarrow B_{(g)}$, the data on the right were obtained for moles of A as the reaction proceeded, The avera rate of disappearance of A between 10 and 20 seconds equals:

Time (s)	0.00	10.0	20.0	30.0	40
Moles of A	0.124	0.110	0.088	0.073	0.0

A) 1.9×10^{-3} B) 2.2×10^{-3} C) 3.1×10^{-3} D) 1.75×10^{-3}
E) 1.50×10^{-3}

ln $[N_2O_5]$ and the time, and Δ Ln$[N_2O_5]$ was - 1.16 during a time period of 2030 s. The value of the rate constant k equals:

A) 7.25×10^{-4} B) 8.29×10^{-4} C) 7.73×10^{-4} D) 6.44×10^{-4} E) 5.71×10^{-4}

7. During the experimental determination of the rate law for the reaction A + B → P, it was found that the rate increased by a factor of 9 when the concentration of B was tripled, this reaction is ----------order in B

A) Zero B) first C) second D) third E) forth

8. Determine the rate law expression for the reaction $2A + B_2 + C \rightarrow A_2B + BC$, using the kinetic data provided on the table:

Run	$[A]_\circ$	$[B_2]_\circ$	$[C]_\circ$	Initial rate (M/s)
1	1.2	1.0×10^{-2}	1.0	1.6×10^{6}
2	2.4	1.0×10^{-2}	1.0	1.6×10^{6}
3	1.2	2.0×10^{-2}	1.0	3.2×10^{6}
4	1.2	1.0×10^{-2}	2.0	6.4×10^{6}

A) $R = k[A][C]^2$ B) $R = k[A][B_2]^2$

C) $R = k[B_2]^2[C]^2$ D) $R = [A]^2[B_2]^2$

E) $R = k[B_2][C]^2$

9. The rate constant k for the reaction $CH_3N{\equiv}C \rightarrow CH_3C{\equiv}N$ equals 6.29×10^{-4} s^{-1}, if the $[CH_3N{\equiv}C]_\circ$ is 1.00×10^{-3} M, then its concentration after 3.5×10^{3} s equals:

A) 3.89×10^{-4} B. 1.11×10^{-4} C. 2.84×10^{-4} D. 2.08×10^{-4} E. 5.33×10^{-4}

10. For the second order reaction A → products, 25 % of A were found to be consumed in 60.0. What is the second half life (in hr) when the initial concentration of A equals 1.0 M?

A) 466.2 B) 371.7 C) 600.6 D) 800.0 E) 1136.4

11. The rate constant for the reaction A→ B is 0.039 $M^{-1}s^{-1}$. The final concentration of A after 45 s was 0.30 M, its initial concentration equals:

A) 0.410 B) 0.463 C) 0.392 D) 0.480 E) 0.633

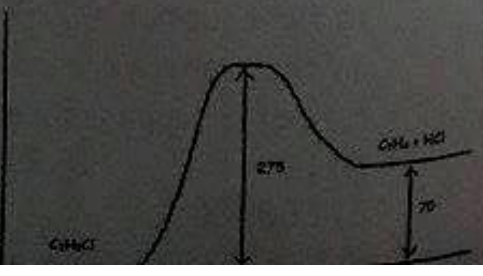
12. The rate constant k for a certain reaction was 3.7×10^{-3} s^{-1} at 25 °C. If the activation energy for this reaction was 43.6 kJ/mol, then the value of rate constant for this reaction at 75 °C equals:

A) 5.7×10^{-2} B) 4.6×10^{-2} C) 3.7×10^{-2} D) 2.4×10^{-2} E) 4.9×10^{-3}

13. The rate equation for the reaction: $A \rightarrow B$ is $R = k[A]^0$. If the initial concentration of the reactant [A] is 1.0 mol L^{-}, the half-life period of the reaction is:

A. k/[A] B. [A]/k C. 2[A]/k D) [A]/2k E) 2k/[A]

14. Carefully inspect the profile on the right for the reaction $C_2H_5Cl \rightarrow C_2H_4 + HCl$; what is the activation energy of the REVERSE reaction, and is the FORWARD reaction exothermic or endothermic?



B) 205 kJ and exothermic reverse reaction
C) 200 kJ and endothermic forward reaction
D) 200 kJ and exothermic reverse reaction
E) 205 kJ and endothermic forward reaction

15. Which one of the following statement is **INCORRECT** concerning a catalyst?
A) A homogeneous catalyst is in the same phase as compared to the reactants.
B) A heterogeneous catalyst is in a different phase as compared to the reactants.
C) A catalyst enhances the rate by decreasing the Ea of the reaction.
D) A catalyst is consumed during the reaction.
E) A catalyst is chemically unchanged.

16. For the reaction $CO_{2(s)} \rightleftharpoons CO_{2(g)}$, if the standard enthalpy of sublimation equals 28.2 $kJ.mol^{-1}$, then the standard entropy of sublimation (in $J.mol^{-1}$) equals:
A) 87.9 B) 233.0 C) 94.6 D) 209.5 E) 81.2

17. Use the data provided on the right to find ΔH for the combustion of 14.4 g of phenol (C_6H_5OH, 94.11 g/mol)
A) - 467.2 B) - 824.0 C) - 438.0 D) - 561.3 E) - 496.4

	$C_6H_5OH_{(s)}$	$+7O_{2(g)} \rightarrow$	$6CO_{2(g)} +$	$3H_2O_{(l)}$
$\Delta H°_f$ (kJ/mol)	- 165.0		- 393.5	- 285.8

18. Use the data provided in the table below to find ΔH (kJ/mol) for the reaction: $N \equiv N + 3\ H\text{-}H \rightarrow 2NH_3$
A) -79 B) 75 C) -109 D) 79 E) -75



Bond	BE (kJ/mol)
N ≡ N	+ 945
H-H	+ 436
N-H	+ 388

19. Which one of the following reactions has a **POSITIVE** entropy change?
A) $BF_{3(g)} + NH_{3(g)} \rightarrow F_3BNH_{3(s)}$
B) $2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$
C) $H_2O_{(g)} \rightarrow H_2O_{(l)}$
D) $2NH_4NO_{3(s)} \rightarrow 2N_{2(g)} + 4H_2O_{(g)} + O_{2(g)}$
E) $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{4(g)}$

For the reaction $2CO_{(g)} + 2H_{2(g)} \rightarrow CO_{2(g)} + CH_{4(g)}$, if $\Delta H° = -247.3$ kJ/mol and $\Delta S° = -256.5$ J/mol K the value of $\Delta G°$ for the reaction:
A) - 170.9 B) 175.6 C) 160.9 D) 181.6 E) 170.9

A solution was prepared by dissolving 50.0 mL of $CHCl_3$ in enough amount of acetone to produce 100 L solution. If the density of $CHCl_3$ was 1.486 g/mL, the molar concentration of this solution equals:
A) 3.73×10^{-3} B) 6.21×10^{-3} C) 7.46×10^{-3} D) 5.60×10^{-3}
E) 8.70×10^{-3}

22. How many nitrogen atoms are in 125.00 g $NH_4H_2PO_4$ (molar mass 114 g/mol)?

A) 2.64×10^{24} B) 3.96×10^{24} C) 4.28×10^{24} D) 2.85×10^{24} E) 6.60×10^{2}

23. Consider the equation: $Ba(OH)_{2(aq)} + H_2SO_{4(aq)} \rightarrow BaSO_{4(s)} + 2H_2O_{(l)}$, if 380.0 mL 0.273 M $Ba(OH)_{2(aq)}$ solution were mixed with 500.0 mL of 0.520 M $H_2SO_{4(aq)}$, 20 g $BaSO_{4(s)}$ (233.4 g/mol) were obtained, the % yield of reaction equals:

A) 67.07 % B) 62.93 % C) 83.60 % D) 75.33 E) 87.73 %

24. Choose the correct set of transition elements:

A) B, Ge, Te, Po B) Yb, Sm, Eu, Dy C) Pa, Np, Am, No

D) P, S, Cl, Br E) Sc, Zr, Cd, Os

The correct set of quantum numbers (n, l, m_l, m_s) for the second electron orbital is:

A) 4, 2, -2, $\frac{1}{2}$ B) 4, 2, 0, $\frac{1}{2}$ C) 4, 1, 0, $\frac{1}{2}$ D) 4, 1, 1, $\frac{1}{2}$ E) 4, 2, -1,

